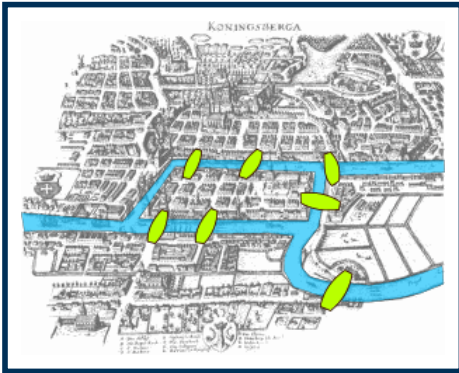


L2: An Introduction



This visualization shows the seven bridges of **Königsberg**. The birth of graph theory took place in 1736 when Leonhard Euler showed that it is not possible to walk through all seven bridges by crossing each of them once and only once.

Food for Thought

Try to model this problem with a graph in which each bridge is represented by an edge, and the landmass at each end of a bridge is represented by a node. The graph should have four nodes (*upper, lower, the island in the middle, and the landmass at the right*) and seven edges. What is the property of this graph that does not allow to walk through each edge once and only once?

You can start from any node you want, and end at any node you want. It is ok to visit the same node multiple times but you should cross each edge only once (*this is referred to as a Eulerian path in graph theory*).